Oriented Power Pooling and Polynomial Root Centering for Diophantine Consistency Statements

Research prepared for Vladimir Reshetnikov and ProveIt
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Abstract

We give a complete construction that combines all seventeen power requests in ProveIt's direct C222 arithmetic encoding into one rational-integrality certificate. Multiplication by the base orients the error terms even when bases have different signs. Zero bases are allowed in the local theorem; the retained arithmetic equations subsequently exclude them. This avoids a circular assumption about the radix and removes nine certificate coordinates. We then apply explicitly listed acyclic unit-pivot substitutions. The resulting fixed integer polynomials include forms in 87 unrestricted integers of degree 8,388,612, in 88 of degree 65,540, in 89 of degree 32,768, and moderate-degree alternatives listed below. Their arithmetic circuits use only integer literals of absolute value at most 256; their expanded coefficients are larger and are bounded separately.

A separate compact result resolves the earlier square-transport question: an inverse Lucas correction permits the transport weight b^2, followed by an integral polynomial root center. It produces a ten-positive-variable polynomial of exact degree 224 in 90 binary operations, together with 85-operation degree-232 companions. This branch retains large calibrated coefficients and has a different domain from the direct integer branch.

For every delivered equation, composition with the specifically identified inherited refutation calibration establishes in ZFC that absence of a solution in its stated domain is equivalent to Con(ZFC). The new unbounded arguments are ordinary mathematical proofs. Independent finite checks authenticate the formulas, witness transformations, degrees, and circuits; no new end-to-end proof-assistant compilation is claimed.

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1 Results, comparison, and the meaning of simplicity

The task is to specify an actual polynomial with fixed integer coefficients, rather than an unspecified instance of a universal representation theorem. The delivered direct family has the following precise property. For every listed pair (n, d) there is a completely specified $P_{n,d} \in \mathbb{Z}[X_0, \ldots, X_{n-1}]$, of exact total degree d, such that

$$\mathrm{ZFC} \vdash \quad (\neg\exists \mathbf{X} \in \mathbb{Z}^n \; P_{n,d}(\mathbf{X}) = 0) \iff \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

Here consistency means syntactic consistency for the effective ZFC presentation used in the retained calibration. None of the construction's integer coefficients depends on deciding whether that theory is consistent.

1.1 The direct family

Integer variables	Exact degree	Residuals	Operations	H
137	12	74	2,548	55
121	16	58	2,502	84
118	22	55	2,523	58
111	32	48	2,502	107
106	38	43	2,457	147
104	64	41	2,451	223
99	128	36	2,483	556
94	256	31	2,468	788
93	328	30	2,448	788
89	32,768	26	2,453	82,436
88	65,540	25	2,450	164,875
87	8,388,612	24	2,447	21,104,473

In the table, H is an integer such that every collected coefficient has absolute value strictly below 2^H. These are proved upper bounds, not exact coefficient heights. Operations count binary additions, subtractions, and multiplications in the supplied circuit, with a square counted as a multiplication and a multiplication by a constant also counted. They are not bit-complexity estimates. All listed variables are independently quantified over $\mathbb{Z}$, and every variable occurs in the actual polynomial;

the independent checker certifies the latter with nonzero modular partial derivatives, beyond mere occurrence in the computation graph.

The 87-variable form is the smallest-dimensional delivered direct form; its degree is substantially higher than the other profiles. It is obtained from the 88-variable form by one additional unit pivot. The 88-variable form has degree 65,540, and the 89-variable form reduces the degree to 32,768 at the cost of one coordinate. The 111-variable degree-32 form is a useful moderate-degree choice, with a comparatively small proved coefficient-height bound. A reader whose priority is short evaluation rather than unrestricted integer variables should instead examine the compact positive-domain companion in Section 7.

1.2 What is improved, and what is inherited

The comparison snapshot is the actual repository commit

`0c48d332df6b0e50d4e5b6251c531898f5682c72`.

This matters because the intake contained newer constructions than the frontier summarized in the older top-level README. The immediate direct parent is package 158, with 225 integer variables and 162 quadratic residuals. Package 168 already supplied a two-base reciprocal certificate and a 215-variable degree-ten companion. Its final discussion suggested oriented weights for more bases. We credit that suggestion. The present contribution is the completed five-base theorem, including temporary zero bases, the simultaneous bootstrap, the actual polynomials, and their composition with certified coordinate elimination.

Package 152 had already applied acyclic monic elimination to a different direct parent. Its relevant reported points were

$$(144, 12),\ (125, 16),\ (117, 32),\ (111, 64),\ (102, 256),\ (94, 65540),$$

where each pair is (unrestricted integer variables, exact degree). Our family improves the dimension at these selected degrees or provides additional tradeoffs; it does not improve every metric simultaneously. For example the supplied factored evaluators can require more binary operations than package 152's evaluators. The comparison is to authenticated repository constructions, not a proof of a global optimum.

The direct route has no native coefficient tuple and does not use a K215 machine simulation, Jones's universal polynomial, or Sun's integer-domain reduction. Its starting point is the repository's numerical proof calculus C222. The compact ten-positive-variable companion retains the separate native/K215 calibration and four large fixed coefficients. These two ancestries and domains are kept explicit throughout.

The existence of Diophantine descriptions for recursively enumerable relations is classical [7, 8]. That general theorem does not identify the best explicit equation for this particular consistency statement. Nor does a universal variable bound, such as the recent ten-integer-variable result [9], settle the simultaneous optimization of degree, coefficient size, circuit size, and calibration complexity.

2 The frozen parent and the exact finite definition

Let $R_0, \dots, R_{161} \in \mathbb{Z}[x_0, \dots, x_{224}]$ be the literal quadratic residuals in Appendix A. The parent polynomial is

$$P_{225} = \sum_{r=0}^{161} R_r^2.$$

Its byte-identical coefficient file has SHA-256

`fee84c408828aba3e7ceebe9a37f58509327063aa7fd078fd49c80adf3af4e56`.

The original Git blob is `425a70a4960e0c60ebedfcae641f08be15dbdcc0`. The parent is not a parameter to be selected later: its complete finite coefficient list is frozen and included.

All newly pooled systems are specified by the same following finite recipe. Delete the ten old coordinates

$$149, 150, 151, 152, 153, 154, 221, 222, 223, 224 \tag{2}$$

and the ten old rows

$$105, 106, 107, 108, 109, 110, 158, 159, 160, 161. \tag{3}$$

Retain the other coordinates in increasing old-index order, relabeling them $u_0, \dots, u_{214}$. Retain the other residuals in increasing old-row order. Append the one, two, or three new integer coordinates and rows specified in Section 5. This gives the raw all-five, staged, and paired systems in respectively 216, 217, and 218 variables.

The difference coordinates $D_i = X_i - AY_i$ are already present in the parent. The entire finite request table is as follows. The symbols X_i are reconstructed as $D_i + Z_i$, where the retained multiplication rows give $Z_i = AY_i$. Set $A = x_{142} + 1$, $q = x_{102} - 1$, $t = x_0$.

Request	Group	Index k_i	Output m_i	D_i	Y_i	Z_i
0	0	x_0	x_5	x_{146}	x_{147}	x_{148}
1	1	x_1	x_{18}	x_{182}	x_{183}	$x_{184} + 1$
2	2	x_1	$x_{102} - 1$	x_{182}	x_{183}	$x_{184} + 1$
3	4	$x_0 + x_3$	x_{103}	x_{155}	x_{156}	x_{157}
4	3	x_0	x_{104}	x_{146}	x_{147}	x_{148}
5	4	$x_9 - 1$	x_{108}	x_{158}	$x_{159} - 1$	x_{160}
7	4	x_{10}	x_{111}	x_{161}	x_{162}	x_{163}
8	4	x_{11}	x_{114}	x_{164}	x_{165}	x_{166}
9	4	x_{12}	x_{117}	x_{167}	x_{168}	x_{169}
10	4	x_{13}	x_{120}	x_{170}	x_{171}	x_{172}
11	3	x_5	x_{121}	x_{173}	x_{174}	x_{175}
12	4	x_{14}	x_{124}	x_{176}	x_{177}	x_{178}
13	4	x_{15}	x_{127}	x_{179}	x_{180}	x_{181}
14	3	x_1	x_{128}	x_{182}	x_{183}	$x_{184} + 1$
15	4	x_{16}	x_{132}	x_{185}	x_{186}	x_{187}
16	4	x_{17}	x_{135}	x_{188}	x_{189}	x_{190}
17	4	x_{100}	x_{138}	x_{191}	x_{192}	x_{193}

For each of the five base groups, define the following polynomial abbreviations, which introduce no new unknowns:

$$\begin{aligned} T_j &= 2An_j - n_j^2 - 1, \\ H_j &= \sum_{i \in I_j} Y_i(m_i - D_i - Z_i), \\ S_j &= \sum_{i \in I_j} \big((m_i - D_i)^2 - Y_i^2\big), \\ E_j &= S_j - 2n_j H_j. \end{aligned} \tag{4}$$

The five bases have request multiplicities $1, 1, 1, 3, 11$, respectively:

$$n_0 = x_3 + 1 = F^2 + 1, \quad n_1 = x_4 = B, \quad n_2 = x_{194} = BH, \quad n_3 = q, \quad n_4 = q + 1,$$

Finally apply the finite pivot lists in Appendix B. In each list the row and variable numbers refer to the indicated raw pooled system, not to the old parent. A pair (r, v) replaces the coordinate u_v by the integer polynomial obtained by solving its unit-linear row $R_r = 0$. Definitions are recursively substituted in their acyclic dependency order. Keep the remaining coordinates in increasing raw-index order, substitute into every unremoved residual, and sum their squares. This is a finite, complete definition of every $P_{n,d}$ in the results table, independently of running the provided software.

The JSON circuits are a more convenient equivalent presentation. Their internal registers are polynomial abbreviations; only their declared input variables are quantified. They contain integer constants and the operations $+, -, \times$, with no divisions, factorials, variable exponents, or unevaluated universal-polynomial parameters.

3 Oriented pooling with temporarily zero bases

3.1 Canonical signed Pell coordinates

For an integer A with $|A| > 1$, define the canonical Pell coordinates by $(X_0, Y_0) = (1, 0)$ and

$$X_{k+1} = AX_k + (A^2 - 1)Y_k, \qquad Y_{k+1} = X_k + AY_k \quad (k \geq 0),$$

and $X_{-k} = X_k$, $Y_{-k} = -Y_k$. These definitions work for either sign of A. They give $X_k^2 - (A^2-1)Y_k^2 = 1$. For $T = 2An - n^2 - 1$ one has $(n - A)^2 \equiv A^2 - 1 \pmod{T}$. For $n \neq 0$ and $k \in \mathbb{Z}$, or for arbitrary n and $k \geq 0$, the Pell group homomorphism therefore gives

$$X_k + (n - A)Y_k \equiv n^k \pmod{T}. \tag{5}$$

When $n \neq 0$, negative exponents in this congruence mean modular inverses. They exist because $T \equiv -1 \pmod{n}$. No rational power occurs in the final polynomial.

Theorem 3.1 (One certificate for all active bases)**.** *Let $W \geq 258$ and $|A| > W^W$ be integers. There are at most five blocks of requests and at most seventeen requests in total. In block j the base is an arbitrary integer n_j. A request consists of integers m_i, k_i and the canonical signed pair (X_{k_i}, Y_{k_i}) at A. Assume*

$$n_j^2 < W, \qquad m_i^2 < W, \qquad k_i^2 < W.$$

Set $D_i = X_{k_i} - AY_{k_i}$, $Z_i = AY_{k_i}$, and define T_j, H_j, S_j, E_j by (4). Then every T_j is nonzero. The polynomial equation

$$\mathcal{R} = \sum_j n_j E_j \prod_{\ell \neq j} T_\ell - z \prod_j T_j = 0 \tag{6}$$

has an integer witness z if and only if every request at a nonzero base is exact: $m_i = n_j^{k_i}$ in $\mathbb{Q}$. When $|n_j| \geq 2$, exactness forces $k_i \geq 0$. A zero-base block contributes zero and is deliberately unconstrained at this stage.

Proof. Put $b = \lfloor \sqrt{W - 1} \rfloor$, so $b \geq 16$ and every absolute base, output, and index is at most b. If $n_j = 0$, then $T_j = -1$. Otherwise

$$|T_j| \geq 2|A||n_j| - n_j^2 - 1 \geq 2|A| - W > W^W, \qquad \operatorname{sgn} \frac{n_j}{T_j} = \operatorname{sgn} A. \tag{7}$$

In particular no denominator vanishes.

Write $\delta_i = m_i - D_i - n_j Y_i$. Direct expansion, using $Z_i = AY_i$, gives the exact identity

$$\sum_{i \in I_j} \delta_i^2 = E_j - T_j \sum_{i \in I_j} Y_i^2. \tag{8}$$

Dividing (6) by the nonzero product of the T_j, and subtracting the integral correction $\sum n_j Y_i^2$, shows that

$$Q_0 = \sum_j \sum_{i \in I_j} \frac{n_j \delta_i^2}{T_j}$$

is an integer. Terms from zero-base blocks vanish.

Let $J = \{j : n_j \neq 0\}$, $s = |J| \leq 5$, and set $C = \prod_{j \in J} n_j^b$, with the empty product equal to 1. This is a nonzero integer. For an active request define

$$d_i = C m_i - \frac{C}{n_j^b} n_j^{b+k_i}.$$

This is an integer: the quotient is an integer, and $b + k_i \geq 0$. Congruence (5) gives $C\delta_i \equiv d_i \pmod{T_j}$. Hence

$$Q_1 = \sum_{j \in J} \sum_{i \in I_j} \frac{n_j d_i^2}{T_j}$$

differs from $C^2 Q_0$ by an integer and is itself an integer. No coprimality between different denominators is used.

If J is empty the conclusion is immediate. Otherwise

$$|C| \leq b^{sb}, \qquad |d_i| \leq b^{sb+1} + b^{(s+1)b} \leq 2b^{6b}.$$

Writing $r \leq 17$ for the number of requests gives

$$\sum_{j \in J} \sum_{i \in I_j} |n_j| d_i^2 \leq 4r b^{12b+1} < W^{6b+2} \leq W^W. \tag{9}$$

For the strict inequality use $4r \leq 68 < W$, $b^2 < W$, and $b < W$. For the last inequality use $6b + 2 \leq b^2 + 1 \leq W$, valid for $b \geq 16$. Equations (7) and (9) imply $|Q_1| < 1$. Thus $Q_1 = 0$. Every nonzero summand has the same sign, so every $d_i = 0$. Division by C in $\mathbb{Q}$ proves $m_i = n_j^{k_i}$. If $|n_j| \geq 2$ and $k_i = -h < 0$, this would imply $m_i n_j^h = 1$, impossible for integers with $h > 0$. Unit bases are handled without incorrectly inferring a sign for their indices.

Conversely, exact powers at the nonzero bases give $T_j \mid \delta_i$ by (5), and then $T_j \mid E_j$ by (8). Therefore

$$z = \sum_{j : n_j \neq 0} \frac{n_j E_j}{T_j} \in \mathbb{Z}$$

satisfies (6). Zero-base terms contribute nothing. □

Remark 3.2 (Why the weights and zero cases matter). Without the factors n_j, terms from bases of different signs can cancel. Merely squaring the denominators would not repair the proof: a congruence modulo T_j gives no corresponding control modulo T_j^2. The factors n_j orient the terms while making the zero-base case harmless. The active set, floor, and variable powers above are proof devices only. Equation (6) is an ordinary polynomial.

4 Recovering the five nonzero bases without circularity

The parent supplies the following frame before interpreting the power requests: $W = 2 + \sum a_\nu^2 \geq 258$, with every actual base, index, and proposed output among the 36 explicitly authenticated affine atoms; $|A| > W^W$; and the canonical signed Pell pairs. Its orientation-first argument uses retained frame,

norm, residue, and cap rows. It does not use the ten deleted power-certificate rows. The exact atom identity and the arithmetic proof dependencies are included in the package.

Use the semantic coordinates

$$F = x_2, \quad t = x_0, \quad V = x_5, \quad B = x_4, \quad N = x_{18}, \quad q = x_{102} - 1, \quad M = x_{128}, \quad BH = x_{194}.$$

The following deductions are valid at every zero of the new system.

First, $R_0 = 0$ gives $x_3 = F^2$, and $R_{139} = 0$ gives $t = 16 + x_{195}^2$. The common remainder row $R_{154} = 0$ has the form

$$M^2 = 1 + 4\left(S^2 + \sum_{i=1}^{11} R_i'^2\right) + a^2 + b^2 + c(c+1). \tag{10}$$

Every summand on the right is nonnegative for integer arguments. Thus $M \neq 0$ and $|S| < |M|/2$. Rows R_{155}, R_{156} say

$$Bv_B = M, \qquad FM = v_B V + S.$$

Consequently $B, v_B \neq 0$. With the integer $s_0 = FB - V$, these equations give $S = v_B s_0$ and

$$|s_0| < |B|/2. \tag{11}$$

None of these conclusions used an interpreted power.

The first base $F^2 + 1$ is positive. Theorem 3.1 therefore already recovers $V = (F^2+1)^t \geq 1$. Put $u = |F|$, $b_0 = |B|$. If $b_0 = 1$, integrality and (11) give $s_0 = 0$, whereas $V = FB \leq u < (u^2+1)^t = V$, a contradiction. If $u \geq 1$ and $b_0 \leq u$, then

$$V = FB - s_0 < (u + \tfrac{1}{2})b_0 \leq u^2 + u/2 < (u^2+1)^2 \leq V,$$

also a contradiction. Thus

$$|B| \geq 2, \qquad |F| < |B|, \qquad V < |B|^2. \tag{12}$$

The last inequality follows from $u \leq b_0 - 1$ and $V < (u + 1/2)b_0$.

The unchanged row $R_{138} = 0$ gives

$$BH = B\big(2(x_9 + x_{10} + \cdots + x_{17}) - 1\big).$$

The factor in parentheses is odd, hence nonzero even before the CRT representatives have been shown nonnegative. Therefore $|BH| \geq 2$. The theorem now recovers $N = B^x$, $q = (BH)^x$, and $x \geq 0$. If $x = 0$, then $q = 1$; if $x > 0$, then $|q| \geq 2$. In both cases $q \neq 0, -1$, so both remaining groups are active. All seventeen exact powers have now been recovered.

The temporary case $x = 0$ is excluded by $M = q^x = 1$, contradicting $Bv_B = M$ and $|B| \geq 2$. The retained ratio for q^t then gives $x \geq t$, so $|q| \geq 2^{16}$; the subsequent factorial, signed remainder, CRT, and Horn-flow interpretation is the parent's unchanged argument. In particular factoriality is not assumed to justify the signs needed for its own derivation.

4.1 Witness maps and the inherited consistency endpoint

For each deleted singleton, restore its modulus T_j and its old quotient $v_j = -\delta_j/T_j$. These are integers on the new zero set by the exact power congruence. For an old same-base pool restore

$$L_j = E_j/T_j, \qquad K_j = 2H_j + (2A - n_j)L_j.$$

These satisfy its original two rows. The independent finite checker proves that the ten deleted coordinates are private to the ten deleted rows. Thus the restoration gives an actual zero of P_{225}.

In the forward direction there is even a polynomial witness map. At an old zero, $\delta_j + T_j v_j = 0$ and

$$E_j/T_j = T_j v_j^2 + Y_j^2$$

for each singleton. The all-five witness can therefore be chosen as

$$z = \sum_{j=0}^{2} n_j (T_j v_j^2 + Y_j^2) + qL_q + (q+1)L_{q+1}.$$

The reverse restoration is integral on the solution set but is not a polynomial automorphism on arbitrary tuples. No bijection of all old certificate witnesses, or preservation of their multiplicities, is claimed.

The inherited logical interfaces are

$$\texttt{KEndpoint.c222_iff_zf_refutation}: \quad \mathrm{C222}(\bot) \iff \mathrm{Ref}_{\mathrm{ZF}}(\bot),$$
$$\texttt{ConstructibleChoice.zf_refutation_iff_zfc_refutation}: \quad \mathrm{Ref}_{\mathrm{ZF}}(\bot) \iff \mathrm{Ref}_{\mathrm{ZFC}}(\bot).$$

The parent arithmetic proof includes its square-shell recoding of the numerical calculus. The logical calibration uses ordinary first-order soundness and completeness and set-model constructions, including an internally defined well-founded part and constructibility. It should not be described as merely a newly implemented primitive-recursive translator of refutations. The inherited audit records why these constructions are formalizable in ZFC and do not assume an externally well-founded model or the consistency statement being represented. Composition gives (1), without deciding either side.

5 Grouping the requests to control degree

The one-pool formula (6) has residual degree 12 in the raw coordinates, hence gives a degree-24 sum of squares in 216 integers. Two alternative groupings reduce degree before coordinate elimination.

Staged grouping. For the first three blocks use

$$\mathcal{R}_{012} = \sum_{j=0}^{2} n_j E_j \prod_{\substack{0 \le \ell \le 2 \\ \ell \ne j}} T_\ell - z_0 T_0 T_1 T_2.$$

For the two consecutive bases use

$$\mathcal{R}_{34} = E_3 T_4 + E_4 T_3 - z_1 T_3 T_4.$$

The first group uses Theorem 3.1. The bootstrap through $q = (BH)^x$ then makes $q, q+1$ nonzero and of the same sign, so T_3, T_4 have the same sign. The unweighted two-base version of the same rational-integrality proof applies, as in package 168. The two residuals have exact degrees 8 and 5. Keeping the other 152 quadratic rows gives 154 residuals in 217 integers and exact degree 16.

Paired grouping. Keep the first singleton quotient but substitute its modulus directly:

$$\mathcal{R}_0 = \delta_0 + z_0 T_0.$$

Use an oriented pair for B, BH and an unweighted pair for $q, q+1$:

$$\mathcal{R}_{12} = n_1 E_1 T_2 + n_2 E_2 T_1 - z_1 T_1 T_2, \qquad \mathcal{R}_{34} = E_3 T_4 + E_4 T_3 - z_2 T_3 T_4.$$

The first equation is just the parent's first power row with its modulus definition substituted. It recovers V at the already positive base F^2+1. The oriented pair then recovers N and q, and the last pair completes the bootstrap. The residual degrees are 3, 6, and 5, giving 155 rows in 218 integers and exact degree 12. The two first nontrivial bases share the same indexed Pell pair, which makes their expanded pair numerator particularly short.

These are three exact presentations of the same existence problem, not an assertion that a higher-degree profile is preferable for every purpose. The coefficient files also retain the expanded individual residuals as an independent definition. We do not claim to have collected every coefficient of the final high-degree sums of squares.

6 Eliminating defined coordinates with exact certificates

Theorem 6.1 (Acyclic unit-pivot elimination)**.** *Let $Q_0,\dots,Q_{m-1}\in\mathbb{Z}[u_0,\dots,u_{N-1}]$. Select distinct rows and distinct coordinates so that each selected row is $Q_r=\varepsilon u_v+f_v$, where $\varepsilon=\pm1$ and u_v does not occur in f_v. Suppose the directed dependency graph of the selected definitions is acyclic. Recursively substitute $u_v=-\varepsilon f_v$ into every unselected row, obtaining $\widehat{Q}_r$ in the remaining n coordinates. Then projection to the remaining coordinates is a bijection from the old simultaneous zero set to the new simultaneous zero set, with an integer polynomial inverse. In particular*

$$\exists\mathbf{u}\in\mathbb{Z}^N\ \sum Q_r(\mathbf{u})^2=0 \quad\longleftrightarrow\quad \exists\mathbf{X}\in\mathbb{Z}^n\ \sum \widehat{Q}_r(\mathbf{X})^2=0.$$

Proof. Use a topological order of the selected definitions. Every removed coordinate is uniquely determined by earlier definitions and the retained coordinates; all coefficients remain integral because the pivot is a unit. The selected rows vanish identically after substitution. Every unselected row becomes precisely its stated substituted polynomial. These facts prove both inverse identities on the corresponding zero sets. A sum of integer squares is zero exactly when each summand is zero. □

The selected lists are finite certificates, so reproduction does not depend on rediscovering them with an optimizer. Some pairs solve a contracted singleton power row for its difference coordinate or output; that is legitimate whenever the explicit unit-pivot test holds. There is no requirement that a row originally called a "power" row be immune to a later purely polynomial elimination.

6.1 Degree, height, and genuine variable occurrence

Assign degree 1 to every retained input. Each eliminated definition gets the maximum weighted degree of its source monomials; each remaining residual gets the same weighted upper bound. Let D be the maximum. To prove that this bound is exact, evaluate the homogeneous component of degree D of a residual after substituting the deterministic input direction $a_v=v+3$, modulo $p=1,000,000,007$. The supplied certificate records a nonzero result. An integer polynomial with a nonzero reduction under one such specialization is not the zero polynomial. Thus a residual has degree exactly D. Over $\mathbb{R}$, the sum of squares of the highest homogeneous components cannot cancel, so the single polynomial has exact degree $2D$. The proof does not rely on treating a circuit's syntactic degree as automatically exact.

For coefficient bounds, recursively assign every input coefficient norm 1. A definition $\sum c_m\prod u_i$ receives the bound $\sum|c_m|\prod L_i$ for its coefficient ℓ^1 norm. If L_r bounds the norm of the substituted residual, then $\sum_r L_r^2$ bounds the norm, hence the height, of the final polynomial. Its bit length is the table's strict exponent H.

Finally, circuit reachability alone cannot exclude cancellation of an input from the actual polynomial. The independent verifier differentiates the complete circuit modulo p and records an evaluation at which each partial derivative is nonzero. Those certificates prove actual dependence on every declared

input. They are algebraic nonvanishing witnesses, not probabilistic evidence about the existence of zeros.

7 A compact positive-domain companion

This companion has ten strictly positive integer variables. Its domain must be kept separate from the unrestricted-integer equations in the main construction. It combines the unsquared strong trace of the pinned `163-trace-radical` contribution with the inverse-Lucas correction of `164-unsquared-factors-linear-selectors` [4, 5]. A new square transport weight then permits a polynomial root center and lowers the exact degree.

Circuit	Variables	Degree	Binary operations
Trace parent	10	232	87
Inverse correction with trace	10	232	85
Square transport with trace	10	232	85
Centered companion	10	224	90

The degree 224 point is a measured degree/cost tradeoff. The claimed result is neither a global minimum nor an assertion of literature-wide priority. This compact branch retains its inherited native/K215 calibration. Its ordinary proof is separate from its finite algebraic checker.

7.1 Entire polynomial

The independent variables are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The four fixed coefficients β, K_0, M_C, M_F are the whole indexed BFS943/radix67 tuple, with $\beta = 2^{590069} - 1$. Their complete finite recipe, 436-character seed, numerical payloads, and independent reconstruction program are included. All raw and compressed hashes agree with the pinned parent. The largest port has 592,884 bits; the four raw payloads occupy 295,388 bytes. Every symbol below is a polynomial abbreviation, not another quantified variable:

$$\begin{gathered}
h = \beta J, \quad q = h + 1, \quad C_0 = q - F - \alpha, \\
X = qw, \quad Y = qs, \quad k = \eta + \zeta, \quad c = kY + \eta, \\
A = Y(X + 1) + 1, \quad \Delta = A^2 - 1, \quad H = 4A - 5, \\
v = XY^2, \quad \ell = vk = (XY)(kY), \quad E = 1 + \ell(\ell + k), \\
R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
B = (K_0 + w)C_0 - F, \quad U = ic^2, \quad a = \Delta U^2 E, \\
F_0 = 1 + a, \quad f = 1 + 2a, \quad G = EF_0, \quad Q = 4\Delta a F_0, \\
b = cT - R, \quad V = fb - c, \quad K = X(3c - 2X) + 2, \\
e_6 = 6(k - R - 1) - \ell(k^2 - 4), \quad D = 6q^6, \quad M = hDH.
\end{gathered}$$

For a weight L, set

$$Z_0 = hD\big[H(n - LB) + K\big] - H\big[DB + h(e_6 - 6q^4 s)\big].$$

The centered companion uses

$$L = b^2, \qquad C = AUE, \qquad S = CV, \qquad Z = Z_0 - MS$$

and the single equation

$$\boxed{P = Q(Z^2 - M^2GV^2) + (M^2G - Z^2) = 0.} \tag{13}$$

The two 85-operation controls use $Z = Z_0$ and respectively $L = bV$ or $L = b^2$. All three literal DAGs are supplied. Fixed powers are multiplication gates; no variable exponent, recurrence, floor, or unspecified coefficient occurs in the polynomial.

For proof notation only, put

$$e_* = \frac{e_6}{6}, \quad \rho = \frac{e_*}{q^6}, \quad \sigma = \frac{s}{q^2}, \quad t = \frac{B}{h}, \quad W = 1 + hL, \quad \kappa = \frac{K}{H}, \quad z = \frac{Z}{M}.$$

Without yet asserting integrality, the centered carrier identity is

$$z = n - S + \kappa + \sigma - Wt - \rho. \tag{14}$$

For an uncentered control, omit S.

7.2 The square transport theorem

The root extraction, rational adjacent-Pell descent, signed folding, and third-index exclusions in the trace source do not use the old cubic, the weight, or the carrier center. Before assuming any sign for B, C_0, R, b, V, their outputs are

$$\begin{gathered} G = g^2, \quad E = \tau^2, \quad F_0 = f_0^2, \quad z = \varepsilon g y, \quad QV^2 - (Q-1)y^2 = 1, \\ c = \psi_A(p), \quad p \geq 5 \text{ odd}, \quad k = 2\psi_{1+2v}(m), \quad m \geq 2, \\ c > 8XY^3 \geq 8q^4, \quad c > X, \quad K > 0, \quad b \neq 0, \quad \operatorname{sgn} V = \operatorname{sgn} b, \\ Q > f^2, \quad c < f/2, \quad |V| > 8Q^2, \quad |b| > 8f^3 - \tfrac{1}{2}. \end{gathered}$$

Here g, τ, f_0, y are positive integers and $\varepsilon \in \{1, -1\}$. The sequence ψ_A has initial values 0,1 and recurrence $\psi_A(j+2) = 2A\psi_A(j+1) - \psi_A(j)$. If the auxiliary index were 3, its two sign cases would give, respectively,

$$f(b - 4\Delta f) = c - (4\Delta + 3), \qquad f(b + 4\Delta f) = c + (4\Delta + 3) :$$

Both right sides lie strictly between 0 and $2c < f$, whereas each left side is a multiple of f, a contradiction.

We now establish the new bound needed for the center. The identities give

$$\frac{a^2}{g^2} = \Delta U^2 \frac{a}{1+a} > 9, \qquad \frac{C^2}{G} < \frac{A^2}{A^2 - 1} < 4.$$

Thus $g < f/6$ and $C < 2g$. Also $y < \frac{3}{2}|V|$ and $|V| < \frac{3}{2}f|b|$. For either root sign,

$$\begin{aligned} |\varepsilon g y + CV| &\leq gy + C|V| < \tfrac{7}{2}g|V| < \tfrac{7}{8}f^2|b| \\ &< \tfrac{7}{32}b^2 < \tfrac{1}{4}b^2. \end{aligned}$$

We used $|b| > 4f^2$, a consequence of the stronger bound above.

The inverse-Lucas correction is nonnegative before rank sign recovery: $vk(k^2 - 4)/6 \geq 0$, because $k \geq 2$. Consequently

$$e_* \leq k - R - 1 < b, \qquad \rho < \frac{|b|}{q^6} < \tfrac{1}{4}b^2.$$

If $B \le -1$, then $Wt < -L$. Solving (14) for positive n gives

$$n < \tfrac{1}{4}L - L + \tfrac{1}{4}L < 0,$$

a contradiction. Therefore $B \ge 0$, and then $C_0 > 0$. This treats all integer ranks R, including negative and zero ranks. The missing adverse-cubic estimate identified in the trace source's open b^2 question is unnecessary for the inverse correction.

7.3 Soundness and the prime 3 ordering

After recovering $C_0 > 0$, the calibrated mask bound gives $3q < R < q^4 - 2$, hence $c > 2R$ and $b, V > 0$. Since

$$f - 1 = 2\Delta i^2 c^4 E, \qquad V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c},$$

the same normalized-rank theorem gives $p = R$. In particular R is positive odd, q is even, and J is odd. The new center S is an integer polynomial, so integrality of z implies integrality of $z + S$. The remaining carrier argument is therefore the one in the inverse-Lucas source; the weight enters only through $W \equiv 1 \pmod{h}$.

For clarity, its essential arithmetic is the recurrence expansion

$$k = 2m + 8vC_m + 32v^2 Z_m, \qquad C_m = \binom{m+1}{3}, \qquad Z_m \in \mathbb{Z}_{\ge 0}.$$

Writing $d = 2m - R - 1$ and $A_m = C_m + 4vZ_m$, one obtains

$$3(e_* - d) = 4v^2\big[24Z_m - A_m\{k^2 + 2mk + (2m)^2 - 4\}\big].$$

The integer k is even, so $k(k^2 - 4)/6$ and hence e_* are integers. Carrier integrality gives $q^6 \mid e_* - q^4 s$, hence $q^4 \mid e_*$. Prime-by-prime valuations of the last identity give $q^4 \mid e_* - d$, including the possible loss of one factor 3. The bound $1 \le m < R < q^4$ forces $d = 0$.

One must not yet conclude $q^6 \mid e_*$. First the whole-product projection from $H \mid 6hK$, with its retained strict size bound, gives $X = 2^R$. Since $q \mid X$, q is a power of 2. Only then can the factor 3 be canceled and $q^6 \mid e_*$ recovered. The remaining congruences give $q^2 \mid s$, $h \mid B$, and integral positive κ. Thus

$$X = 2^R, \quad q^3 \mid Y, \quad t = B/h \in \mathbb{Z}_{\ge 0}, \quad (K_0 + w)C_0 - F = t(q - 1).$$

The inherited whole-tuple native/refutation calibration now yields a finite syntactic ZFC refutation. This step retains the entire calibrated tuple, not merely its elementary coefficient inequalities.

7.4 A cofinal positive lift of the centered equation

Conversely a finite refutation supplies the canonical outer witnesses in the two pinned source theorems. In particular q is a power of 2, $R \equiv 3 \pmod 4$, $X = 2^R$, $q^3 \mid Y$, $C_0 > 0$, and $t \ge 0$, while ρ, σ, κ are integers. Their exact Pell-ratio interval supplies positive η, ζ. Choose the strong seed with $c^2\tau \mid I_0$, so $i = I_0/(c^2\tau) > 0$, $g = \tau f_0$, and $f = 2f_0^2 - 1$.

The retained adjacent-Pell completeness theorem supplies positive V_*, y such that $V_* + c + Rf^2$ is divisible by cf^2. The unsquared-trace coordinate

$$T = \frac{V_* + c + Rf}{cf}$$

is a positive integer, because $c \mid f - 1$. For an uncentered control take $n = gy + Wt + \rho - \kappa - \sigma$. The inherited bounds $|\rho| < g/3$ and $\kappa + \sigma < g/2$ prove positivity.

For the centered equation the negative root is useful. Put $u_0 = U^2E \geq 2$. The exact identity

$$(Q-1)C^2 - QG = E[Q(u_0-1) - A^2u_0]$$

is positive: $Q > 4\Delta^2 u_0$ and $4\Delta^2 > A^2$. Therefore $C > g\sqrt{Q/(Q-1)}$. If $D_0 = (Q-1)C^2 - QG$, then D_0 is a positive integer and

$$CV_* - gy > \frac{D_0 V_*}{2C(Q-1)} \geq \frac{V_*}{2C(Q-1)}.$$

There are cofinally many permissible V_*. The matrix

$$\begin{pmatrix} 2Q-1 & 2(Q-1) \\ 2Q & 2Q-1 \end{pmatrix}$$

preserves the adjacent Pell norm, has determinant 1, and strictly increases positive pairs. A power of it is the identity modulo cf^2; iterating that power preserves the required congruence and makes V_* unbounded. No additional polynomial variable is used.

Choose $V_* > 2C(Q-1)(\kappa + \sigma + |\rho|)$ and set

$$n = CV_* - gy + (1+hb^2)t + \rho - \kappa - \sigma.$$

Then n is a strictly positive integer, the centered carrier is $z = -gy$, and (13) vanishes. This proves positive zero-existence equivalence; it does not mistake an arbitrary integer shear for an automorphism of the positive domain.

7.5 Exact degree and independent finite verification

The cancellation is an identity over a free polynomial ring:

$$\begin{aligned} P = {} & (Q-1)Z_0^2 - 2(Q-1)MCVZ_0 \\ & + M^2V^2[QE(U^2E-1) - C^2] + M^2G. \end{aligned}$$

The degrees of Q, M, V, U, E, Z_0, C, G are respectively $80, 11, 40, 7, 14, 21, 25, 50$. Its unique degree 224 term is $QM^2V^2U^2E^2$; all other summands have strictly lower degree. With $q_0 = \beta J$, $k_0 = \eta + \zeta$, and $B_0 = k_0 sT - q_0^2(q_0 - F)$, the complete highest homogeneous part is

$$[P]_{224} = 9\,2^{10} q_0^{96} w^{24} s^{56} i^{10} k_0^{32} B_0^2.$$

The coefficient of $J^{96}w^{24}s^{58}i^{10}\eta^{34}T^2$ is $9\,2^{10}\beta^{96} > 0$, proving the exact degree on the authentic coefficient slice. The uncentered controls have complete top

$$[P]_{232} = -9\,2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_0^2.$$

The checker imports no producer. It verifies every actual DAG gate by independently specified polynomial cuts, all live inputs and coefficient ports, the exact cancellation identity, and the complete leading forms. Both alternative operators at every gate are deliberately introduced and rejected:180 controls for the centered circuit and170 for each control. Normal and optimized Python runs pass. This authenticates the finite artifacts; the unbounded mathematical arguments above remain essential.

8 Verification, reproducibility, and proof boundaries

The deliverable has four distinct layers of evidence.

1. The unbounded pooling theorem, its zero-base bootstrap, the compact transport estimate, and the cofinal positive lift are mathematical proofs. The independent internal reviews separately check these changed arguments.

2. The frozen parent, full request table, and literal pool expansions are authenticated with exact integer arithmetic. The checker reconstructs all seventeen request energies and the retained Pell and multiplication rows, checks coordinate privacy, and verifies every output coefficient.

3. The projection checker imports no producer or optimizer. It independently checks unit pivots, acyclicity, every local circuit block, the complete source-row partition, factored pool formulas, the final sum of squares, exact degrees, coefficient bounds, and input dependence. Corrupted artifacts are deliberately rejected.

4. The compact checker verifies all polynomial circuit gates through independent cuts, the root-center cancellation, the complete leading forms, and every fixed coefficient payload. Its constructor reproduces the four coefficient integers from the printed finite seed and recipe.

The retained parent theorem and the named C222/ZF/ZFC logical calibration are inherited mathematical dependencies. This package does not contain a new transitive Lean or Rocq build, does not emit an encoded object-language ZFC proof, and does not prove Con(ZFC). Independent AI-assisted reviews are not external human peer review. The ordinary equivalence proof and the finite algebraic verification should be evaluated at their respective scopes.

All core replay programs use Python's standard library. From the package root run

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B code/build_projections.py --check
```

The first command verifies the saved deliverable without performing a root search. The evaluator accepts an explicit tuple and computes its polynomial value or its residuals; it is not a consistency decision procedure. The chosen pivot lists are frozen certificates; rebuilding and verifying them does not require an optimizer or repeat the exploratory search.

The article can be rebuilt with two runs of pdfLaTeX. The exact mathematical definitions are the source coefficient files, pivot certificates, factored circuits, and printed appendices; PDF bytes can vary with the TeX installation. Source pins, attribution, licenses, and a release checksum manifest accompany the package.

9 Further research questions

Choose the best partition of the bases. The all-five, staged, and paired formulas illustrate a genuine degree versus coordinate tradeoff. The best grouping can depend on which semantic coordinates a later monic elimination replaces. An optimizer that jointly chooses groups and polynomial definitions could improve on choosing the grouping first. Its output should remain a small exact certificate.

Compress the signed Pell frame itself. The present theorem shares power-congruence certificates while retaining the parent's canonical-index frame. The next substantial saving would compress those index and orientation witnesses without assuming a desired sign or rank prematurely. A new shared identity should be tested first against negative indices, unit bases, and temporary zero bases.

Exploit stronger height slack. The proof only uses $W^{6b+2} \le W^W$ with $b \ge 16$, leaving a large margin. Additional bounded arithmetic tests might fit into the same rational energy. A proposal must preserve the common sign and justify how every modulus remains nonzero, rather than merely multiplying more congruences together.

Lower the degree of the smallest direct forms. The highest degrees in the 87-, 88-, and 89-variable forms come from long product and squaring chains, not solely from the five-base numerator. Introducing a few strategically selected intermediates, or changing the factorization of the Horn-flow certificate, may lower degree at a better coordinate cost. None of the present pivot lists is claimed optimal.

Shorten the centered compact evaluator. The new degree-224 carrier uses five additional operations compared with the 85-operation controls. A different schedule for the root center, or a different polynomial approximation to $\sqrt{G}$, could improve that tradeoff. Positive completeness must be proved for any new center; an arbitrary integer shear need not preserve positive witnesses.

Formalize the new interface. A useful focused formalization target is Theorem 3.1 with an explicit list of bases and requests, followed by its literal 17-request instance. This would make future groupings reuse a single signed, zero-base-aware theorem. The complete consistency statement would then compose with the repository's existing formal calibration interfaces.

References

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[2] ProveIt, *168-reciprocal-pooling*, `proofs/RECIPROCAL_POOLING.md`, same snapshot. https://github.com/VladimirReshetnikov/ProveIt/tree/0c48d332df6b0e50d4e5b6251c531898f5682c72/Computability/DiophantineConsistency/con-zfc-diophantine-forms/168-reciprocal-pooling.

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A Every residual of the authenticated parent

All variables $x_0, \dots, x_{224}$ in this appendix range over unrestricted integers. The polynomials below are literally the rows of the frozen parent; no omitted scalar parameter occurs. Each expression defines R_r, and the parent equation is $\sum_{r=0}^{161} R_r^2 = 0$. The main text gives the exact coordinate and row deletion lists, the complete request table, and the new pooling rows. These together specify the three raw systems used by the next appendix.

$$R_0 = -x_2^2 + x_3$$

$$R_1 = x_0 - x_0x_5 - x_5 + x_8$$

$$R_2 = -1 + x_1 - x_2 - x_5x_6$$

$$R_3 = -1 + x_1 - x_7x_8$$

$$R_4 = 1 + x_3 - x_4x_{19} + x_{17}$$

$$R_5 = -1 + 2x_9 - x_9^2 - 2x_9x_{10} + 2x_{10} - x_{10}^2 + x_{20}$$

$$\begin{aligned} R_6 = {} & -5 - 30x_9 - 36x_9^2 - 96x_9x_{10} - 24x_9x_{20} - 40x_{10} - 64x_{10}^2 - 32x_{10}x_{20} \\ & - 10x_{20} - 4x_{20}^2 + x_{21} \end{aligned}$$

$$R_7 = -2 + 2x_9 - x_9^2 - 4x_9x_{21} + 4x_{21} - 4x_{21}^2 + x_{22}$$

$$\begin{aligned} R_8 = {} & 1 + 6x_9 - 36x_9^2 - 192x_9x_{21} - 24x_9x_{22} + 16x_{21} - 256x_{21}^2 - 64x_{21}x_{22} \\ & + 2x_{22} - 4x_{22}^2 + x_{23} \end{aligned}$$

$$R_9 = -2 - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{24}$$

$$\begin{aligned} R_{10} = {} & -9 + 7x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{24} + 12x_{10} - 4x_{10}^2 - 8x_{10}x_{24} \\ & + 12x_{24} - 4x_{24}^2 + x_{25} \end{aligned}$$

$$R_{11} = -1 + 2x_9 - x_9^2 - 2x_9x_{11} + 2x_{11} - x_{11}^2 + x_{26}$$

$$\begin{aligned} R_{12} = {} & -6x_9 - 144x_9^2 - 192x_9x_{10} - 192x_9x_{11} - 48x_9x_{20} - 48x_9x_{26} - 8x_{10} - 64x_{10}^2 \\ & - 128x_{10}x_{11} - 32x_{10}x_{20} - 32x_{10}x_{26} - 64x_{11}^2 - 32x_{11}x_{20} - 32x_{11}x_{26} - 2x_{20} - 4x_{20}^2 \\ & - 8x_{20}x_{26} - 4x_{26}^2 + x_{27} \end{aligned}$$

$$R_{13} = -4x_{25}^2 - 8x_{25}x_{27} - 4x_{27}^2 + x_{28}$$

$$\begin{aligned} R_{14} = {} & -6 + 16x_{11} - 16x_{11}^2 - 16x_{11}x_{24} - 16x_{11}x_{26} + 8x_{24} - 4x_{24}^2 - 8x_{24}x_{26} \\ & + 8x_{26} - 4x_{26}^2 + x_{29} \end{aligned}$$

$$R_{15} = 1 + 6x_9 - 36x_9^2 - 96x_9x_{10} - 48x_9x_{11} - 24x_9x_{20} - 48x_9x_{26} - 24x_9x_{29} + 8x_{10} - 64x_{10}^2 - 64x_{10}x_{11} - 32x_{10}x_{20} - 64x_{10}x_{26} - 32x_{10}x_{29} + 8x_{11} - 16x_{11}^2 - 16x_{11}x_{20} - 32x_{11}x_{26} - 16x_{11}x_{29} + 2x_{20} - 4x_{20}^2 - 16x_{20}x_{26} - 8x_{20}x_{29} + 8x_{26} - 16x_{26}^2 - 16x_{26}x_{29} + 4x_{29} - 4x_{29}^2 + x_{30}$$

$$R_{16} = -10 - 5x_9 - x_9^2 - 12x_9x_{10} - 16x_9x_{11} - 4x_9x_{24} - 36x_{10} - 36x_{10}^2 - 96x_{10}x_{11} - 24x_{10}x_{24} - 48x_{11} - 64x_{11}^2 - 32x_{11}x_{24} - 12x_{24} - 4x_{24}^2 + x_{31}$$

$$R_{17} = -2 - 24x_9 - 36x_9^2 - 12x_9x_{10} - 96x_9x_{11} - 24x_9x_{26} - 3x_{10} - x_{10}^2 - 16x_{10}x_{11} - 4x_{10}x_{26} - 32x_{11} - 64x_{11}^2 - 32x_{11}x_{26} - 8x_{26} - 4x_{26}^2 + x_{32}$$

$$R_{18} = 2 - 4x_{31}^2 - 8x_{31}x_{32} - 4x_{32}^2 + x_{33}$$

$$R_{19} = -12 + 16x_9 - 4x_9^2 - 8x_9x_{11} - 8x_9x_{20} - 8x_9x_{24} + 16x_{11} - 4x_{11}^2 - 8x_{11}x_{20} - 8x_{11}x_{24} + 16x_{20} - 4x_{20}^2 - 8x_{20}x_{24} + 16x_{24} - 4x_{24}^2 + x_{34}$$

$$R_{20} = -3 - 24x_9 - 64x_9^2 - 160x_9x_{11} - 64x_9x_{24} - 32x_9x_{26} - 32x_9x_{34} - 26x_{11} - 100x_{11}^2 - 80x_{11}x_{24} - 40x_{11}x_{26} - 40x_{11}x_{34} - 8x_{24} - 16x_{24}^2 - 16x_{24}x_{26} - 16x_{24}x_{34} - 6x_{26} - 4x_{26}^2 - 8x_{26}x_{34} - 4x_{34} - 4x_{34}^2 + x_{35}$$

$$R_{21} = -1 - 18x_9 - 36x_9^2 - 96x_9x_{10} - 24x_9x_{20} - 24x_{10} - 64x_{10}^2 - 32x_{10}x_{20} - 6x_{20} - 4x_{20}^2 + x_{36}$$

$$R_{22} = -2 + 2x_9 - x_9^2 - 4x_9x_{36} + 4x_{36} - 4x_{36}^2 + x_{37}$$

$$R_{23} = 1 + 6x_9 - 36x_9^2 - 192x_9x_{36} - 24x_9x_{37} + 16x_{36} - 256x_{36}^2 - 64x_{36}x_{37} + 2x_{37} - 4x_{37}^2 + x_{38}$$

$$R_{24} = -x_{10}^2 - 4x_{10}x_{38} - 4x_{38}^2 + x_{39}$$

$$R_{25} = -6 - 30x_9 - 64x_9^2 - 128x_9x_{10} - 64x_9x_{20} - 32x_{10} - 64x_{10}^2 - 64x_{10}x_{20} - 14x_{20} - 16x_{20}^2 + x_{40}$$

$$R_{26} = -4 + 8x_{10} - 64x_{10}^2 - 96x_{10}x_{11} - 32x_{10}x_{24} - 32x_{10}x_{40} + 6x_{11} - 36x_{11}^2 - 24x_{11}x_{24} - 24x_{11}x_{40} + 2x_{24} - 4x_{24}^2 - 8x_{24}x_{40} + 4x_{40} - 4x_{40}^2 + x_{41}$$

$$R_{27} = 8x_9 - 64x_9^2 - 96x_9x_{11} - 32x_9x_{26} - 32x_9x_{41} + 6x_{11} - 36x_{11}^2 - 24x_{11}x_{26} - 24x_{11}x_{41} + 2x_{26} - 4x_{26}^2 - 8x_{26}x_{41} + 4x_{41} - 4x_{41}^2 + x_{42}$$

$$R_{28} = -x_{11}^2 - 2x_{11}x_{12} - x_{12}^2 + x_{43}$$

$$R_{29} = 2x_9 - x_9^2 - 4x_9x_{11} - 4x_9x_{43} + 4x_{11} - 4x_{11}^2 - 8x_{11}x_{43} + 4x_{43} - 4x_{43}^2 + x_{44}$$

$$R_{30} = -35 + 24x_9 - 4x_9^2 - 8x_9x_{10} - 8x_9x_{24} - 8x_9x_{26} + 24x_{10} - 4x_{10}^2 - 8x_{10}x_{24} - 8x_{10}x_{26} + 24x_{24} - 4x_{24}^2 - 8x_{24}x_{26} + 24x_{26} - 4x_{26}^2 + x_{45}$$

$$R_{31} = -16x_9^2 - 32x_9x_{11} - 32x_9x_{26} - 32x_9x_{43} - 16x_9x_{44} - 16x_9x_{45} - 4x_{11} - 16x_{11}^2 - 32x_{11}x_{26} - 32x_{11}x_{43} - 16x_{11}x_{44} - 16x_{11}x_{45} - 16x_{26}^2 - 32x_{26}x_{43} - 16x_{26}x_{44} - 16x_{26}x_{45} - 4x_{43} - 16x_{43}^2 - 16x_{43}x_{44} - 16x_{43}x_{45} - 2x_{44} - 4x_{44}^2 - 8x_{44}x_{45} - 4x_{45}^2 + x_{46}$$

$$R_{32} = x_{11} - 25x_{11}^2 - 40x_{11}x_{26} - 20x_{11}x_{29} - 16x_{26}^2 - 16x_{26}x_{29} - 4x_{29}^2 + x_{47}$$

$$R_{33} = -1 + 8x_{10} - 4x_{10}^2 - 8x_{10}x_{24} + 8x_{24} - 4x_{24}^2 + x_{48}$$

$$R_{34} = -1 - 16x_{10}^2 - 8x_{10}x_{11} - 32x_{10}x_{24} - 16x_{10}x_{48} - x_{11}^2 - 8x_{11}x_{24} - 4x_{11}x_{48} - 16x_{24}^2 - 16x_{24}x_{48} - 4x_{48}^2 + x_{49}$$

$$R_{35} = 1 + x_{10} - x_{10}^2 - 4x_{10}x_{11} - 4x_{10}x_{49} - 4x_{11}^2 - 8x_{11}x_{49} - 4x_{49}^2 + x_{50}$$

$$R_{36} = -1 - 4x_{50} - 4x_{50}^2 + x_{51}$$

$$R_{37} = -1 - x_{11}^2 - 8x_{11}x_{50} - 4x_{11}x_{51} - 16x_{50}^2 - 16x_{50}x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{38} = 2 + x_{10} - x_{10}^2 - 4x_{10}x_{11} - 4x_{10}x_{52} - 4x_{11}^2 - 8x_{11}x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{39} = -4x_{47}^2 - 8x_{47}x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{40} = -9 + 7x_9 - x_9^2 - 8x_9x_{47} - 4x_9x_{54} + 24x_{47} - 16x_{47}^2 - 16x_{47}x_{54} + 12x_{54} - 4x_{54}^2 + x_{55}$$

$$R_{41} = -1 - 4x_{55} - 4x_{55}^2 + x_{56}$$

$$R_{42} = -x_{11}^2 - 8x_{11}x_{55} - 4x_{11}x_{56} - 16x_{55}^2 - 16x_{55}x_{56} - 4x_{56}^2 + x_{57}$$

$$R_{43} = -x_9 - x_9^2 - 4x_9x_{11} - 4x_9x_{57} - 4x_{11} - 4x_{11}^2 - 8x_{11}x_{57} - 4x_{57} - 4x_{57}^2 + x_{58}$$

$$R_{44} = -10 - 12x_{58} - 4x_{58}^2 + x_{59}$$

$$R_{45} = 1 + 4x_{10} - 4x_{10}^2 - 8x_{10}x_{24} - 16x_{10}x_{58} - 8x_{10}x_{59} + 4x_{24} - 4x_{24}^2 - 16x_{24}x_{58} - 8x_{24}x_{59} + 8x_{58} - 16x_{58}^2 - 16x_{58}x_{59} + 4x_{59} - 4x_{59}^2 + x_{60}$$

$$R_{46} = x_{10} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{60} - 16x_{24}^2 - 16x_{24}x_{60} - 4x_{60}^2 + x_{61}$$

$$R_{47} = -10 - 12x_{53} - 4x_{53}^2 + x_{62}$$

$$R_{48} = 1 - 16x_{53}^2 - 16x_{53}x_{61} - 16x_{53}x_{62} - 4x_{61}^2 - 8x_{61}x_{62} - 4x_{62}^2 + x_{63}$$

$$R_{49} = x_{11} - x_{11}^2 - 8x_{11}x_{61} - 4x_{11}x_{63} - 16x_{61}^2 - 16x_{61}x_{63} - 4x_{63}^2 + x_{64}$$

$$R_{50} = -4x_9 - 4x_9^2 - 4x_9x_{11} - 8x_9x_{20} - x_{11} - x_{11}^2 - 4x_{11}x_{20} - 4x_{20} - 4x_{20}^2 + x_{65}$$

$$R_{51} = 1 + 4x_9 - 4x_9^2 - 8x_9x_{20} - 8x_9x_{65} + 4x_{20} - 4x_{20}^2 - 8x_{20}x_{65} + 4x_{65} - 4x_{65}^2 + x_{66}$$

$$R_{52} = -2 - 4x_9 - 16x_9^2 - 32x_9x_{20} - 16x_9x_{64} - 16x_9x_{66} - 4x_{20} - 16x_{20}^2 - 16x_{20}x_{64} - 16x_{20}x_{66} - 4x_{64}^2 - 8x_{64}x_{66} - 2x_{66} - 4x_{66}^2 + x_{67}$$

$$R_{53} = -2 - 4x_{67} - 4x_{67}^2 + x_{68}$$

$$R_{54} = -x_2x_{14} + x_{69}$$

$$R_{55} = x_{13} - x_{14} - 2x_{23}x_{69} - x_{70}$$

$$R_{56} = x_{13} - x_{14} - 4x_{25}x_{69} - 2x_{28}x_{69} - x_{71}$$

$$R_{57} = x_{13} - x_{14} - 2x_{30}x_{69} - 6x_{69} - x_{72}$$

$$R_{58} = x_{13} - x_{14} - 4x_{31}x_{69} - 2x_{33}x_{69} - x_{73}$$

$$R_{59} = x_{13} - x_{14} - 2x_{35}x_{69} + 6x_{69} - x_{74}$$

$$R_{60} = x_{13} - x_{14} - 4x_{38}x_{69} - 2x_{39}x_{69} - x_{75}$$

$$R_{61} = x_{13} - x_{14} - 2x_{42}x_{69} - x_{76}$$

$$R_{62} = x_{13} - x_{14} - 2x_{46}x_{69} - 4x_{69} - x_{77}$$

$$R_{63} = -x_9x_{10} + x_9x_{11} + x_{10} + x_{10}x_{11} - x_{11} - x_{11}^2 + x_{78}$$

$$R_{64} = 1 + 2x_{12} + x_{12}^2 - x_{78}^2 + x_{79}$$

$$R_{65} = -x_{13}^2 + 2x_{13}x_{14} + 2x_{13}x_{81} - x_{14}^2 - 2x_{14}x_{81} - x_{79}^2 + x_{80} - x_{81}^2$$

$$R_{66} = -4x_{67}x_{69} - 2x_{68}x_{69} + 2x_{69} + x_{81}$$

$$R_{67} = -x_{10}x_{69} - x_{14} + x_{82}x_{83} + x_{84}$$

$$R_{68} = -1 + 2x_2 - 2x_2x_9 - 2x_2x_{20} + x_{82}$$

$$R_{69} = -x_9x_{220} - x_{13} + x_{83} + x_{220}$$

$$R_{70} = -6x_9x_{69} - 8x_{10}x_{69} + x_{10}x_{220} + x_{13} - x_{14} - 2x_{20}x_{69} + x_{85}$$

$$R_{71} = -x_{13}x_{70} + x_{14}x_{70} + x_{86}$$

$$R_{72} = -x_{71}x_{86} + x_{87}$$

$$R_{73} = -x_{72}x_{87} + x_{88}$$

$$R_{74} = -x_{73}x_{88} + x_{89}$$

$$R_{75} = -x_{74}x_{89} + x_{90}$$

$$R_{76} = -x_{75}x_{90} + x_{91}$$

$$R_{77} = -x_{76}x_{91} + x_{92}$$

$$R_{78} = -x_{77}x_{92} + x_{93}$$

$$R_{79} = -x_{84}x_{93} + x_{94}$$

$$R_{80} = -x_{85}x_{94} + x_{95}$$

$$R_{81} = -x_{95}^2 + x_{96}$$

$$R_{82} = -x_5 - x_5x_{17} + x_{17} + x_{97}$$

$$R_{83} = -x_{15}x_{97} + x_{98}$$

$$\begin{aligned} R_{84} = {} & -2x_1^2 + 2x_1x_{13} + 2x_1x_{14} + 2x_1x_{98} + 2x_1x_{99} - x_{13}^2 - 2x_{13}x_{98} - x_{14}^2 \\ & - 2x_{14}x_{99} + x_{18}x_{101} - x_{80}x_{96} - x_{98}^2 - x_{99}^2 \end{aligned}$$

$$R_{85} = -x_5x_{16} + x_{16} - x_{16}x_{97} + x_{99}$$

$$\begin{aligned} R_{86} = {} & -9x_9 - 9x_{10} - 4x_{20} - 2x_{23} - 4x_{25} - 2x_{28} - 2x_{30} - 4x_{31} \\ & - 2x_{33} - 2x_{35} - 4x_{38} - 2x_{39} - 2x_{42} - 2x_{46} - 4x_{67} - 2x_{68} \\ & + x_{100} \end{aligned}$$

$$R_{87} = -x_{102}x_{105} + x_{105} + x_{107}$$

$$R_{88} = -x_4x_{128} + x_{103}x_{210} - x_{106} - x_{107}x_{128}$$

$$R_{89} = x_{108}x_{210} - x_{109}x_{131} - x_{110}$$

$$R_{90} = x_{111}x_{210} - x_{112}x_{131} - x_{113}$$

$$R_{91} = x_{114}x_{210} - x_{115}x_{131} - x_{116}$$

$$R_{92} = x_{117}x_{210} - x_{118}x_{131} - x_{119}$$

$$R_{93} = x_{120}x_{211} - x_{122}x_{131} - x_{123}$$

$$R_{94} = x_{124}x_{211} - x_{125}x_{131} - x_{126}$$

$$R_{95} = -x_{18}x_{128} + x_{131}$$

$$R_{96} = x_{127} - x_{129}x_{131} - x_{130}$$

$$R_{97} = -x_{131}x_{133} + x_{132} - x_{134}$$
$$R_{98} = -x_{131}x_{136} + x_{135}x_{210} - x_{137}$$
$$R_{99} = -x_{131}x_{139} + x_{138}x_{210} - x_{140}$$
$$R_{100} = x_{141} - 2x_{141}x_{143} + x_{144}$$
$$R_{101} = -x_{141}x_{144} - x_{144} + x_{145}$$
$$R_{102} = 2x_{142} + x_{142}^2 + x_{144}^2 - x_{145}^2$$
$$R_{103} = -x_{142}x_{147} - x_{147} + x_{148}$$
$$R_{104} = -1 + x_{146}^2 + 2x_{146}x_{148} + x_{147}^2$$
$$R_{105} = x_3^2 - 2x_3x_{142} - 2x_{142} + x_{149}$$
$$R_{106} = -x_3x_{147} + x_5 - x_{146} - x_{147} + x_{149}x_{150}$$
$$R_{107} = 1 - 2x_4 + x_4^2 - 2x_4x_{142} + x_{151}$$
$$R_{108} = -x_4x_{183} + x_{18} + x_{151}x_{152} - x_{182}$$
$$R_{109} = -1 + 2x_{142}x_{194} - x_{153} + 2x_{194} - x_{194}^2$$
$$R_{110} = -1 + x_{102} + x_{153}x_{154} - x_{182} - x_{183}x_{194}$$
$$R_{111} = -x_{142}x_{156} - x_{156} + x_{157}$$
$$R_{112} = -1 + x_{155}^2 + 2x_{155}x_{157} + x_{156}^2$$
$$R_{113} = 1 + x_{142} - x_{142}x_{159} - x_{159} + x_{160}$$
$$R_{114} = x_{158}^2 + 2x_{158}x_{160} - 2x_{159} + x_{159}^2$$
$$R_{115} = -x_{142}x_{162} - x_{162} + x_{163}$$
$$R_{116} = -1 + x_{161}^2 + 2x_{161}x_{163} + x_{162}^2$$
$$R_{117} = -x_{142}x_{165} - x_{165} + x_{166}$$
$$R_{118} = -1 + x_{164}^2 + 2x_{164}x_{166} + x_{165}^2$$
$$R_{119} = -x_{142}x_{168} - x_{168} + x_{169}$$
$$R_{120} = -1 + x_{167}^2 + 2x_{167}x_{169} + x_{168}^2$$
$$R_{121} = -x_{142}x_{171} - x_{171} + x_{172}$$
$$R_{122} = -1 + x_{170}^2 + 2x_{170}x_{172} + x_{171}^2$$
$$R_{123} = -x_{142}x_{174} - x_{174} + x_{175}$$
$$R_{124} = -1 + x_{173}^2 + 2x_{173}x_{175} + x_{174}^2$$
$$R_{125} = -x_{142}x_{177} - x_{177} + x_{178}$$
$$R_{126} = -1 + x_{176}^2 + 2x_{176}x_{178} + x_{177}^2$$
$$R_{127} = -x_{142}x_{180} - x_{180} + x_{181}$$
$$R_{128} = -1 + x_{179}^2 + 2x_{179}x_{181} + x_{180}^2$$
$$R_{129} = 1 - x_{142}x_{183} - x_{183} + x_{184}$$
$$R_{130} = -1 + 2x_{182} + x_{182}^2 + 2x_{182}x_{184} + x_{183}^2$$
$$R_{131} = -x_{142}x_{186} - x_{186} + x_{187}$$
$$R_{132} = -1 + x_{185}^2 + 2x_{185}x_{187} + x_{186}^2$$

$$R_{133} = -x_{142}x_{189} - x_{189} + x_{190}$$

$$R_{134} = -1 + x_{188}^2 + 2x_{188}x_{190} + x_{189}^2$$

$$R_{135} = -x_{142}x_{192} - x_{192} + x_{193}$$

$$R_{136} = -1 + x_{191}^2 + 2x_{191}x_{193} + x_{192}^2$$

$$\begin{aligned} R_{137} = &-6 - 2x_0 - 3x_0^2 - 2x_0x_3 - x_1^2 - 2x_3 - 2x_3^2 - x_4^2 \\ &- 2x_5^2 + 2x_9 - x_9^2 - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 \\ &- x_{15}^2 - x_{16}^2 - x_{17}^2 - x_{18}^2 - x_{100}^2 + 2x_{102} - 2x_{102}^2 - x_{103}^2 \\ &- x_{104}^2 - x_{108}^2 - x_{111}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{124}^2 \\ &- x_{127}^2 - x_{128}^2 - x_{132}^2 - x_{135}^2 - x_{138}^2 + x_{141} - x_{194}^2 \end{aligned}$$

$$\begin{aligned} R_{138} = &\, x_4 - 2x_4x_9 - 2x_4x_{10} - 2x_4x_{11} - 2x_4x_{12} - 2x_4x_{13} - 2x_4x_{14} - 2x_4x_{15} \\ &- 2x_4x_{16} - 2x_4x_{17} + x_{194} \end{aligned}$$

$$R_{139} = -16 + x_0 - x_{195}^2$$

$$R_{140} = -x_{142}x_{197} - x_{197} + x_{206}$$

$$R_{141} = -1 + x_{196}^2 + x_{197}^2 - x_{206}^2$$

$$R_{142} = -x_{199} - x_{199}x_{200} + x_{207}$$

$$R_{143} = -1 + x_{198}^2 + x_{199}^2 - x_{207}^2$$

$$R_{144} = -x_{192}^2 + x_{208}$$

$$R_{145} = x_{197} - 2x_{202}x_{208} - x_{208}$$

$$R_{146} = -4x_{197} - 8x_{197}x_{201} + x_{200}$$

$$R_{147} = -x_{142} - x_{196}x_{203} + x_{200}$$

$$R_{148} = -x_{191} - x_{193} - x_{196}x_{204} + x_{198}$$

$$R_{149} = -x_{100} - 4x_{192}x_{205} + x_{199}$$

$$\begin{aligned} R_{150} = &\, 26 + 2x_0^2 + 2x_0x_3 - 2x_0x_{147} - 2x_0x_{156} + x_1^2 - 2x_1x_{183} + x_3^2 \\ &- 2x_3x_{156} + x_5^2 - 2x_5x_{174} + x_9^2 - 2x_9x_{159} + x_{10}^2 - 2x_{10}x_{162} + x_{11}^2 \\ &- 2x_{11}x_{165} + x_{12}^2 - 2x_{12}x_{168} + x_{13}^2 - 2x_{13}x_{171} + x_{14}^2 - 2x_{14}x_{177} + x_{15}^2 \\ &- 2x_{15}x_{180} + x_{16}^2 - 2x_{16}x_{186} + x_{17}^2 - 2x_{17}x_{189} + x_{100}^2 - 2x_{100}x_{192} - x_{142}x_{209} \\ &- 2x_{146} - 2x_{148} + x_{148}^2 - 2x_{155} - 2x_{157} + x_{157}^2 - 2x_{158} + 2x_{159} \\ &- 2x_{160} + x_{160}^2 - 2x_{161} - 2x_{163} + x_{163}^2 - 2x_{164} - 2x_{166} + x_{166}^2 \\ &- 2x_{167} - 2x_{169} + x_{169}^2 - 2x_{170} - 2x_{172} + x_{172}^2 - 2x_{173} - 2x_{175} \\ &+ x_{175}^2 - 2x_{176} - 2x_{178} + x_{178}^2 - 2x_{179} - 2x_{181} + x_{181}^2 - 2x_{182} \\ &+ x_{184}^2 - 2x_{185} - 2x_{187} + x_{187}^2 - 2x_{188} - 2x_{190} + x_{190}^2 - 2x_{191} \\ &- 2x_{193} + x_{193}^2 \end{aligned}$$

$$R_{151} = -x_{104}x_{210} + x_{128}$$

$$R_{152} = -x_{121}x_{211} + x_{128}$$

$$\begin{aligned} R_{153} = &-2 - x_{100}^2 - x_{147}^2 - x_{156}^2 + 2x_{159} - x_{159}^2 - x_{162}^2 - x_{165}^2 \\ &- x_{168}^2 - x_{171}^2 - x_{174}^2 - x_{177}^2 - x_{180}^2 - x_{183}^2 - x_{186}^2 - x_{189}^2 \\ &+ x_{192}^2 - x_{212}^2 - x_{213}^2 - x_{214} - x_{214}^2 \end{aligned}$$

$$
\begin{aligned}
R_{154} = {} & -1 - 4x_{106}^2 - 4x_{110}^2 - 4x_{113}^2 - 4x_{116}^2 - 4x_{119}^2 - 4x_{123}^2 - 4x_{126}^2 \\
& + x_{128}^2 - 4x_{130}^2 - 4x_{134}^2 - 4x_{137}^2 - 4x_{140}^2 - x_{215}^2 - x_{216}^2 - x_{217} \\
& - x_{217}^2 - 4x_{219}^2
\end{aligned}
$$

$$
R_{155} = -x_4x_{218} + x_{128}
$$

$$
R_{156} = x_2x_{128} - x_5x_{218} - x_{219}
$$

$$
R_{157} = -x_2x_{13} + x_{220}
$$

$$
\begin{aligned}
R_{158} = {} & x_{102}x_{222} - 2x_{103}x_{156} + 2x_{108} - 2x_{108}x_{159} - 2x_{111}x_{162} - 2x_{114}x_{165} - 2x_{117}x_{168} \\
& - 2x_{120}x_{171} - 2x_{124}x_{177} - 2x_{127}x_{180} - 2x_{132}x_{186} - 2x_{135}x_{189} - 2x_{138}x_{192} - 2x_{142}x_{222} \\
& + 2x_{155}x_{156} + 2x_{156}x_{157} - 2x_{158} + 2x_{158}x_{159} + 2x_{159}x_{160} - 2x_{160} + 2x_{161}x_{162} + 2x_{162}x_{163} \\
& + 2x_{164}x_{165} + 2x_{165}x_{166} + 2x_{167}x_{168} + 2x_{168}x_{169} + 2x_{170}x_{171} + 2x_{171}x_{172} + 2x_{176}x_{177} \\
& + 2x_{177}x_{178} + 2x_{179}x_{180} + 2x_{180}x_{181} + 2x_{185}x_{186} + 2x_{186}x_{187} + 2x_{188}x_{189} + 2x_{189}x_{190} \\
& + 2x_{191}x_{192} + 2x_{192}x_{193} + x_{221} - 2x_{222}
\end{aligned}
$$

$$
\begin{aligned}
R_{159} = {} & -1 - x_{102}x_{221} + x_{103}^2 - 2x_{103}x_{155} + x_{108}^2 - 2x_{108}x_{158} + x_{111}^2 - 2x_{111}x_{161} \\
& + x_{114}^2 - 2x_{114}x_{164} + x_{117}^2 - 2x_{117}x_{167} + x_{120}^2 - 2x_{120}x_{170} + x_{124}^2 - 2x_{124}x_{176} \\
& + x_{127}^2 - 2x_{127}x_{179} + x_{132}^2 - 2x_{132}x_{185} + x_{135}^2 - 2x_{135}x_{188} + x_{138}^2 - 2x_{138}x_{191} \\
& + x_{155}^2 - x_{156}^2 + x_{158}^2 + 2x_{159} - x_{159}^2 + x_{161}^2 - x_{162}^2 + x_{164}^2 \\
& - x_{165}^2 + x_{167}^2 - x_{168}^2 + x_{170}^2 - x_{171}^2 + x_{176}^2 - x_{177}^2 + x_{179}^2 \\
& - x_{180}^2 + x_{185}^2 - x_{186}^2 + x_{188}^2 - x_{189}^2 + x_{191}^2 - x_{192}^2 + x_{222}
\end{aligned}
$$

$$
\begin{aligned}
R_{160} = {} & x_{102}x_{224} - 2x_{104}x_{147} - 2x_{121}x_{174} - 2x_{128}x_{183} - 2x_{142}x_{224} + 2x_{146}x_{147} + 2x_{147}x_{148} \\
& + 2x_{173}x_{174} + 2x_{174}x_{175} + 2x_{182}x_{183} + 2x_{183} + 2x_{183}x_{184} + x_{223} - 3x_{224}
\end{aligned}
$$

$$
\begin{aligned}
R_{161} = {} & -x_{102}x_{223} + x_{104}^2 - 2x_{104}x_{146} + x_{121}^2 - 2x_{121}x_{173} + x_{128}^2 - 2x_{128}x_{182} + x_{146}^2 \\
& - x_{147}^2 + x_{173}^2 - x_{174}^2 + x_{182}^2 - x_{183}^2 + x_{223} + x_{224}
\end{aligned}
$$

B Complete unit-pivot certificates

For each profile, the following list contains every selected pair (r, v) of raw-system row index and raw-system variable index. The variable order and row order of each raw system are specified in Section 2. Solve each selected row for its listed variable, recursively substitute in an acyclic order, remove those rows and variables, retain all other coordinates in increasing order, and sum the squares of the remaining substituted residuals. There are no existential variables for intermediate definitions. The circuit files implement exactly this recipe, with the pooling numerators factored.

137 integers, degree 12; paired grouping

$(131, 141)$	$(15, 30)$	$(31, 46)$	$(20, 35)$	$(12, 27)$	$(27, 42)$
$(17, 32)$	$(45, 60)$	$(19, 34)$	$(26, 41)$	$(30, 45)$	$(16, 31)$
$(52, 67)$	$(34, 49)$	$(29, 44)$	$(8, 23)$	$(65, 80)$	$(10, 25)$
$(23, 38)$	$(50, 65)$	$(70, 85)$	$(37, 52)$	$(40, 55)$	$(48, 63)$
$(63, 78)$	$(6, 21)$	$(14, 29)$	$(13, 28)$	$(18, 33)$	$(56, 71)$
$(58, 73)$	$(60, 75)$	$(68, 82)$	$(88, 106)$	$(67, 84)$	$(57, 72)$
$(59, 74)$	$(62, 77)$	$(24, 39)$	$(1, 8)$	$(69, 83)$	$(61, 76)$
$(82, 97)$	$(89, 110)$	$(90, 113)$	$(91, 116)$	$(92, 119)$	$(93, 123)$
$(94, 126)$	$(98, 137)$	$(99, 140)$	$(142, 192)$	$(150, 213)$	$(81, 96)$
$(96, 130)$	$(97, 134)$	$(140, 194)$	$(134, 200)$	$(87, 107)$	$(138, 202)$
$(101, 145)$	$(136, 201)$	$(53, 68)$	$(151, 214)$	$(55, 70)$	$(85, 99)$

(83, 98)	(105, 151)	(109, 157)	(111, 160)	(113, 163)	(115, 166)
(119, 172)	(121, 175)	(125, 181)	(127, 184)	(129, 187)	(103, 148)
(117, 169)	(107, 154)	(95, 131)			

121 integers, degree 16; paired grouping

(46, 61)	(70, 85)	(107, 154)	(143, 100)	(10, 25)	(119, 172)
(92, 119)	(100, 144)	(54, 69)	(60, 75)	(42, 57)	(87, 107)
(134, 200)	(8, 23)	(61, 76)	(36, 51)	(4, 17)	(68, 82)
(81, 96)	(33, 48)	(149, 128)	(18, 33)	(142, 192)	(152, 146)
(96, 130)	(88, 106)	(94, 126)	(16, 31)	(113, 163)	(1, 8)
(19, 34)	(40, 55)	(138, 202)	(117, 169)	(97, 134)	(49, 64)
(83, 98)	(51, 66)	(22, 37)	(91, 116)	(25, 40)	(56, 71)
(121, 175)	(111, 160)	(73, 88)	(75, 90)	(59, 74)	(3, 1)
(50, 65)	(101, 145)	(62, 77)	(109, 157)	(71, 86)	(95, 131)
(44, 59)	(129, 187)	(66, 81)	(67, 84)	(123, 178)	(141, 194)
(57, 13)	(82, 97)	(30, 45)	(7, 22)	(38, 53)	(28, 43)
(11, 26)	(35, 50)	(127, 184)	(136, 201)	(29, 44)	(24, 39)
(150, 213)	(64, 79)	(53, 68)	(105, 151)	(115, 166)	(103, 148)
(63, 78)	(99, 140)	(21, 36)	(90, 113)	(77, 92)	(133, 0)
(139, 191)	(93, 123)	(151, 214)	(69, 83)	(98, 137)	(125, 181)
(89, 110)	(12, 27)	(85, 99)	(26, 41)	(47, 62)	(58, 73)
(14, 29)					

118 integers, degree 22; staged grouping

(151, 214)	(138, 202)	(95, 131)	(83, 98)	(81, 96)	(79, 94)
(77, 92)	(75, 90)	(73, 88)	(54, 69)	(0, 3)	(145, 128)
(72, 87)	(78, 93)	(74, 89)	(80, 95)	(136, 201)	(134, 200)
(129, 187)	(127, 184)	(125, 181)	(121, 175)	(119, 172)	(117, 169)
(115, 166)	(113, 163)	(111, 160)	(109, 157)	(105, 151)	(103, 148)
(101, 145)	(87, 107)	(143, 100)	(3, 1)	(133, 0)	(150, 213)
(140, 194)	(100, 144)	(99, 140)	(98, 137)	(97, 134)	(96, 130)
(94, 126)	(93, 123)	(92, 119)	(91, 116)	(90, 113)	(89, 110)
(71, 86)	(139, 191)	(142, 192)	(123, 178)	(85, 99)	(53, 68)
(47, 62)	(44, 59)	(41, 56)	(39, 54)	(36, 51)	(28, 43)
(24, 39)	(13, 28)	(4, 17)	(69, 83)	(88, 106)	(66, 81)
(1, 8)	(82, 97)	(55, 70)	(61, 13)	(107, 154)	(68, 82)
(64, 79)	(18, 33)	(9, 24)	(22, 37)	(11, 26)	(7, 22)
(5, 20)	(63, 78)	(42, 57)	(33, 48)	(49, 64)	(37, 52)
(50, 65)	(29, 44)	(25, 40)	(14, 29)	(23, 38)	(10, 25)
(45, 60)	(30, 45)	(19, 34)	(17, 32)	(16, 31)	(12, 27)
(20, 35)	(31, 46)	(15, 30)			

111 integers, degree 32; staged grouping

(151, 214)	(138, 202)	(95, 131)	(83, 98)	(81, 96)	(79, 94)
(77, 92)	(75, 90)	(73, 88)	(54, 69)	(0, 3)	(145, 128)
(72, 87)	(78, 93)	(74, 89)	(80, 95)	(136, 201)	(134, 200)

(129, 187)	(127, 184)	(125, 181)	(121, 175)	(119, 172)	(117, 169)
(115, 166)	(113, 163)	(111, 160)	(109, 157)	(105, 151)	(103, 148)
(101, 145)	(87, 107)	(143, 100)	(3, 1)	(133, 0)	(150, 213)
(140, 194)	(100, 144)	(99, 140)	(98, 137)	(97, 134)	(96, 130)
(94, 126)	(93, 123)	(92, 119)	(91, 116)	(90, 113)	(89, 110)
(71, 86)	(139, 191)	(142, 192)	(123, 178)	(85, 99)	(53, 68)
(47, 62)	(44, 59)	(41, 56)	(39, 54)	(36, 51)	(28, 43)
(24, 39)	(13, 28)	(4, 17)	(69, 83)	(88, 106)	(66, 81)
(1, 8)	(82, 97)	(55, 70)	(61, 76)	(107, 154)	(68, 82)
(64, 79)	(18, 33)	(9, 24)	(60, 75)	(59, 74)	(57, 72)
(22, 37)	(11, 26)	(7, 22)	(5, 20)	(63, 78)	(42, 57)
(33, 48)	(49, 64)	(37, 52)	(48, 63)	(32, 47)	(35, 50)
(50, 65)	(29, 44)	(132, 188)	(25, 40)	(51, 66)	(21, 36)
(10, 25)	(6, 21)	(45, 60)	(30, 45)	(19, 34)	(26, 41)
(17, 32)	(16, 31)	(12, 27)	(31, 46)		

106 integers, degree 38; paired grouping

(131, 141)	(15, 30)	(31, 46)	(20, 35)	(12, 27)	(27, 42)
(17, 32)	(45, 60)	(19, 34)	(26, 41)	(30, 45)	(16, 31)
(52, 67)	(34, 49)	(29, 44)	(8, 23)	(65, 80)	(10, 25)
(23, 38)	(50, 65)	(70, 85)	(37, 52)	(40, 55)	(48, 63)
(63, 78)	(6, 21)	(14, 29)	(25, 40)	(13, 28)	(18, 33)
(56, 71)	(58, 73)	(60, 75)	(68, 82)	(88, 106)	(67, 84)
(43, 58)	(57, 72)	(59, 74)	(62, 77)	(24, 39)	(7, 22)
(1, 8)	(71, 86)	(69, 83)	(61, 76)	(55, 70)	(82, 97)
(89, 110)	(90, 113)	(91, 116)	(92, 119)	(93, 123)	(94, 126)
(98, 137)	(99, 140)	(142, 192)	(150, 213)	(66, 81)	(64, 79)
(81, 96)	(96, 130)	(97, 134)	(140, 194)	(134, 200)	(87, 107)
(79, 94)	(73, 88)	(75, 90)	(53, 68)	(77, 92)	(21, 36)
(138, 202)	(101, 145)	(136, 201)	(33, 48)	(151, 214)	(143, 193)
(85, 99)	(28, 43)	(100, 144)	(83, 98)	(36, 51)	(47, 62)
(2, 2)	(105, 151)	(109, 157)	(111, 160)	(113, 163)	(115, 166)
(119, 172)	(121, 175)	(125, 181)	(127, 184)	(129, 187)	(41, 56)
(103, 148)	(117, 169)	(139, 191)	(107, 154)	(44, 59)	(123, 178)
(133, 0)	(4, 3)	(39, 54)	(95, 131)	(11, 26)	(9, 24)
(145, 128)	(152, 146)	(132, 188)	(141, 142)		

104 integers, degree 64; paired grouping

(133, 0)	(2, 1)	(0, 3)	(152, 5)	(1, 8)	(55, 13)
(67, 14)	(4, 17)	(5, 20)	(6, 21)	(7, 22)	(9, 24)
(10, 25)	(11, 26)	(12, 27)	(13, 28)	(14, 29)	(15, 30)
(16, 31)	(17, 32)	(18, 33)	(19, 34)	(20, 35)	(21, 36)
(22, 37)	(23, 38)	(25, 40)	(26, 41)	(27, 42)	(28, 43)
(29, 44)	(30, 45)	(31, 46)	(32, 47)	(33, 48)	(34, 49)
(35, 50)	(38, 53)	(39, 54)	(41, 56)	(42, 57)	(43, 58)
(44, 59)	(46, 61)	(47, 62)	(48, 63)	(49, 64)	(50, 65)
(51, 66)	(53, 68)	(56, 71)	(57, 72)	(58, 73)	(59, 74)
(60, 75)	(61, 76)	(62, 77)	(63, 78)	(64, 79)	(65, 80)
(66, 81)	(68, 82)	(70, 85)	(71, 86)	(72, 87)	(73, 88)

(75, 90)	(77, 92)	(78, 93)	(79, 94)	(81, 96)	(82, 97)
(83, 98)	(85, 99)	(86, 100)	(88, 106)	(87, 107)	(89, 110)
(90, 113)	(91, 116)	(92, 119)	(93, 123)	(94, 126)	(96, 127)
(145, 128)	(95, 131)	(97, 132)	(98, 137)	(99, 140)	(141, 142)
(100, 144)	(101, 145)	(103, 148)	(105, 151)	(107, 154)	(109, 157)
(111, 160)	(113, 163)	(115, 166)	(117, 169)	(119, 172)	(121, 175)
(123, 178)	(125, 181)	(127, 184)	(142, 185)	(129, 187)	(132, 188)
(139, 191)	(143, 193)	(140, 194)	(134, 200)	(138, 202)	(150, 213)

99 integers, degree 128; all5 grouping

(131, 141)	(15, 30)	(31, 46)	(20, 35)	(12, 27)	(27, 42)
(17, 32)	(45, 60)	(19, 34)	(26, 41)	(30, 45)	(16, 31)
(52, 67)	(34, 49)	(29, 44)	(8, 23)	(65, 80)	(10, 25)
(23, 38)	(50, 65)	(70, 85)	(37, 52)	(40, 55)	(48, 63)
(63, 78)	(6, 21)	(14, 29)	(25, 40)	(13, 28)	(18, 33)
(56, 71)	(58, 73)	(60, 75)	(68, 82)	(88, 106)	(67, 84)
(43, 58)	(57, 72)	(59, 74)	(62, 77)	(24, 39)	(7, 22)
(1, 8)	(71, 86)	(69, 83)	(61, 76)	(55, 70)	(82, 97)
(89, 110)	(90, 113)	(91, 116)	(92, 119)	(93, 123)	(94, 126)
(98, 137)	(99, 140)	(142, 192)	(150, 213)	(66, 81)	(64, 79)
(81, 96)	(96, 130)	(97, 134)	(140, 194)	(134, 200)	(87, 107)
(79, 94)	(73, 88)	(75, 90)	(53, 68)	(77, 92)	(21, 36)
(74, 89)	(78, 93)	(72, 87)	(76, 91)	(138, 202)	(101, 145)
(136, 201)	(33, 48)	(151, 214)	(143, 193)	(85, 99)	(100, 144)
(83, 98)	(36, 51)	(47, 62)	(2, 2)	(41, 56)	(139, 191)
(44, 59)	(133, 0)	(4, 17)	(39, 54)	(95, 131)	(51, 66)
(35, 50)	(32, 47)	(46, 61)	(105, 151)	(109, 157)	(111, 160)
(113, 163)	(115, 166)	(119, 172)	(121, 175)	(125, 181)	(127, 184)
(129, 187)	(117, 169)	(123, 178)	(107, 154)	(103, 148)	(145, 128)
(132, 188)	(141, 142)	(0, 3)			

94 integers, degree 256; all5 grouping

(131, 141)	(15, 30)	(31, 46)	(20, 35)	(12, 27)	(27, 42)
(17, 32)	(45, 60)	(19, 34)	(26, 41)	(30, 45)	(16, 31)
(52, 67)	(34, 49)	(29, 44)	(8, 23)	(65, 80)	(10, 25)
(23, 38)	(50, 65)	(70, 85)	(37, 52)	(40, 55)	(48, 63)
(63, 78)	(6, 21)	(14, 29)	(25, 40)	(13, 28)	(18, 33)
(56, 71)	(58, 73)	(60, 75)	(68, 82)	(88, 106)	(67, 84)
(43, 58)	(57, 72)	(59, 74)	(62, 77)	(24, 39)	(7, 22)
(1, 8)	(71, 86)	(69, 83)	(61, 76)	(55, 70)	(82, 97)
(89, 110)	(90, 113)	(91, 116)	(92, 119)	(93, 123)	(94, 126)
(98, 137)	(99, 140)	(142, 192)	(150, 213)	(66, 81)	(64, 79)
(81, 96)	(96, 130)	(97, 134)	(140, 194)	(134, 200)	(87, 107)
(79, 94)	(73, 88)	(75, 90)	(53, 68)	(77, 92)	(21, 36)
(74, 89)	(78, 93)	(72, 87)	(76, 91)	(138, 202)	(101, 145)
(136, 201)	(33, 48)	(151, 214)	(143, 193)	(85, 99)	(28, 43)
(100, 144)	(83, 98)	(36, 51)	(47, 62)	(2, 2)	(41, 56)
(22, 37)	(139, 191)	(44, 59)	(133, 0)	(4, 17)	(54, 69)
(39, 54)	(95, 131)	(51, 66)	(11, 26)	(9, 24)	(49, 64)

$(105, 151)$	$(109, 157)$	$(111, 160)$	$(113, 163)$	$(115, 166)$	$(119, 172)$
$(121, 175)$	$(125, 181)$	$(127, 184)$	$(129, 187)$	$(35, 50)$	$(117, 169)$
$(32, 47)$	$(123, 178)$	$(107, 154)$	$(103, 148)$	$(145, 128)$	$(132, 188)$
$(141, 142)$	$(0, 3)$				

93 integers, degree 328; staged grouping

$(131, 141)$	$(15, 30)$	$(31, 46)$	$(20, 35)$	$(12, 27)$	$(27, 42)$
$(17, 32)$	$(45, 60)$	$(19, 34)$	$(26, 41)$	$(30, 45)$	$(16, 31)$
$(52, 67)$	$(34, 49)$	$(29, 44)$	$(8, 23)$	$(65, 80)$	$(10, 25)$
$(23, 38)$	$(50, 65)$	$(70, 85)$	$(37, 52)$	$(40, 55)$	$(48, 63)$
$(63, 78)$	$(6, 21)$	$(14, 29)$	$(25, 40)$	$(13, 28)$	$(18, 33)$
$(56, 71)$	$(58, 73)$	$(60, 75)$	$(68, 82)$	$(88, 106)$	$(67, 84)$
$(43, 58)$	$(57, 72)$	$(59, 74)$	$(62, 77)$	$(24, 39)$	$(7, 22)$
$(1, 8)$	$(71, 86)$	$(69, 83)$	$(61, 76)$	$(55, 70)$	$(82, 97)$
$(89, 110)$	$(90, 113)$	$(91, 116)$	$(92, 119)$	$(93, 123)$	$(94, 126)$
$(98, 137)$	$(99, 140)$	$(142, 192)$	$(150, 213)$	$(66, 81)$	$(64, 79)$
$(81, 96)$	$(96, 130)$	$(97, 134)$	$(140, 194)$	$(134, 200)$	$(87, 107)$
$(79, 94)$	$(73, 88)$	$(75, 90)$	$(53, 68)$	$(77, 92)$	$(21, 36)$
$(74, 89)$	$(78, 93)$	$(72, 87)$	$(76, 91)$	$(138, 202)$	$(101, 145)$
$(136, 201)$	$(33, 48)$	$(151, 214)$	$(143, 193)$	$(85, 99)$	$(28, 43)$
$(100, 144)$	$(83, 98)$	$(36, 51)$	$(47, 62)$	$(2, 2)$	$(105, 151)$
$(109, 157)$	$(111, 160)$	$(113, 163)$	$(115, 166)$	$(119, 172)$	$(121, 175)$
$(125, 181)$	$(127, 184)$	$(129, 187)$	$(41, 56)$	$(117, 169)$	$(22, 37)$
$(139, 191)$	$(107, 154)$	$(44, 59)$	$(133, 0)$	$(4, 17)$	$(54, 69)$
$(39, 54)$	$(95, 131)$	$(51, 66)$	$(123, 178)$	$(11, 26)$	$(9, 24)$
$(49, 64)$	$(145, 128)$	$(103, 148)$	$(5, 20)$	$(3, 1)$	$(35, 50)$
$(32, 47)$	$(132, 188)$	$(141, 142)$	$(0, 3)$		

89 integers, degree 32,768; all5 grouping

$(151, 214)$	$(138, 202)$	$(95, 131)$	$(83, 98)$	$(81, 96)$	$(79, 94)$
$(77, 92)$	$(75, 90)$	$(73, 88)$	$(54, 69)$	$(0, 3)$	$(145, 128)$
$(72, 87)$	$(78, 93)$	$(74, 89)$	$(80, 95)$	$(76, 91)$	$(136, 201)$
$(134, 200)$	$(129, 187)$	$(127, 184)$	$(125, 181)$	$(121, 175)$	$(119, 172)$
$(117, 169)$	$(115, 166)$	$(113, 163)$	$(111, 160)$	$(109, 157)$	$(105, 151)$
$(103, 148)$	$(101, 145)$	$(87, 107)$	$(143, 100)$	$(3, 1)$	$(133, 0)$
$(150, 213)$	$(140, 194)$	$(100, 144)$	$(99, 140)$	$(98, 137)$	$(97, 134)$
$(96, 130)$	$(94, 126)$	$(93, 123)$	$(92, 119)$	$(91, 116)$	$(90, 113)$
$(89, 110)$	$(139, 191)$	$(71, 86)$	$(141, 142)$	$(85, 99)$	$(53, 68)$
$(47, 62)$	$(44, 59)$	$(41, 56)$	$(39, 54)$	$(36, 51)$	$(28, 43)$
$(24, 39)$	$(13, 28)$	$(4, 17)$	$(69, 83)$	$(88, 106)$	$(66, 81)$
$(142, 192)$	$(123, 178)$	$(1, 8)$	$(82, 97)$	$(61, 76)$	$(67, 84)$
$(55, 70)$	$(2, 2)$	$(64, 79)$	$(18, 33)$	$(9, 24)$	$(107, 154)$
$(68, 82)$	$(62, 77)$	$(59, 74)$	$(60, 75)$	$(57, 72)$	$(58, 73)$
$(56, 71)$	$(22, 37)$	$(11, 26)$	$(7, 22)$	$(63, 78)$	$(42, 57)$
$(33, 48)$	$(5, 20)$	$(70, 85)$	$(49, 64)$	$(37, 52)$	$(48, 63)$
$(32, 47)$	$(65, 80)$	$(46, 61)$	$(35, 50)$	$(38, 53)$	$(50, 65)$
$(29, 44)$	$(43, 58)$	$(25, 40)$	$(51, 66)$	$(14, 29)$	$(132, 188)$
$(21, 36)$	$(6, 21)$	$(10, 25)$	$(8, 23)$	$(23, 38)$	$(40, 55)$
$(34, 49)$	$(27, 42)$	$(52, 67)$	$(30, 45)$	$(19, 34)$	$(17, 32)$

$(16, 31)$ $(26, 41)$ $(12, 27)$ $(20, 35)$ $(31, 46)$ $(15, 30)$
$(131, 141)$

88 integers, degree 65,540; all5 grouping

$(131, 141)$ $(15, 30)$ $(31, 46)$ $(20, 35)$ $(12, 27)$ $(27, 42)$
$(17, 32)$ $(45, 60)$ $(19, 34)$ $(26, 41)$ $(30, 45)$ $(16, 31)$
$(52, 67)$ $(34, 49)$ $(29, 44)$ $(8, 23)$ $(65, 80)$ $(10, 25)$
$(23, 38)$ $(50, 65)$ $(70, 85)$ $(37, 52)$ $(40, 55)$ $(48, 63)$
$(63, 78)$ $(6, 21)$ $(14, 29)$ $(25, 40)$ $(13, 28)$ $(18, 33)$
$(56, 71)$ $(58, 73)$ $(60, 75)$ $(68, 82)$ $(88, 106)$ $(67, 84)$
$(43, 58)$ $(57, 72)$ $(59, 74)$ $(62, 77)$ $(24, 39)$ $(7, 22)$
$(1, 8)$ $(71, 86)$ $(69, 83)$ $(61, 76)$ $(55, 70)$ $(82, 97)$
$(89, 110)$ $(90, 113)$ $(91, 116)$ $(92, 119)$ $(93, 123)$ $(94, 126)$
$(98, 137)$ $(99, 140)$ $(142, 192)$ $(150, 213)$ $(66, 81)$ $(64, 79)$
$(81, 96)$ $(96, 130)$ $(97, 134)$ $(140, 194)$ $(134, 200)$ $(87, 107)$
$(79, 94)$ $(73, 88)$ $(75, 90)$ $(53, 68)$ $(77, 92)$ $(21, 36)$
$(74, 89)$ $(78, 93)$ $(72, 87)$ $(76, 91)$ $(138, 202)$ $(101, 145)$
$(136, 201)$ $(33, 48)$ $(151, 214)$ $(143, 193)$ $(85, 99)$ $(28, 43)$
$(100, 144)$ $(83, 98)$ $(36, 51)$ $(47, 62)$ $(2, 2)$ $(41, 56)$
$(22, 37)$ $(139, 191)$ $(44, 59)$ $(133, 0)$ $(4, 17)$ $(54, 69)$
$(39, 54)$ $(95, 131)$ $(51, 66)$ $(11, 26)$ $(9, 24)$ $(49, 64)$
$(5, 20)$ $(3, 1)$ $(80, 95)$ $(105, 151)$ $(109, 157)$ $(111, 160)$
$(113, 163)$ $(115, 166)$ $(119, 172)$ $(121, 175)$ $(125, 181)$ $(127, 184)$
$(129, 187)$ $(35, 50)$ $(117, 169)$ $(32, 47)$ $(123, 178)$ $(107, 154)$
$(103, 148)$ $(86, 100)$ $(145, 128)$ $(42, 57)$ $(132, 188)$ $(141, 142)$
$(38, 53)$ $(0, 3)$

87 integers, degree 8,388,612; all5 grouping

$(0, 3)$ $(1, 8)$ $(2, 2)$ $(3, 1)$ $(4, 17)$ $(5, 20)$
$(6, 21)$ $(7, 22)$ $(8, 23)$ $(9, 24)$ $(10, 25)$ $(11, 26)$
$(12, 27)$ $(13, 28)$ $(14, 29)$ $(15, 30)$ $(16, 31)$ $(17, 32)$
$(18, 33)$ $(19, 34)$ $(20, 35)$ $(21, 36)$ $(22, 37)$ $(23, 38)$
$(24, 39)$ $(25, 40)$ $(26, 41)$ $(27, 42)$ $(28, 43)$ $(29, 44)$
$(30, 45)$ $(31, 46)$ $(32, 47)$ $(33, 48)$ $(34, 49)$ $(35, 50)$
$(36, 51)$ $(37, 52)$ $(38, 53)$ $(39, 54)$ $(40, 55)$ $(41, 56)$
$(42, 57)$ $(43, 58)$ $(44, 59)$ $(45, 60)$ $(46, 61)$ $(47, 62)$
$(48, 63)$ $(49, 64)$ $(50, 65)$ $(51, 66)$ $(52, 67)$ $(53, 68)$
$(54, 69)$ $(55, 70)$ $(56, 71)$ $(57, 72)$ $(58, 73)$ $(59, 74)$
$(60, 75)$ $(61, 76)$ $(62, 77)$ $(63, 78)$ $(64, 79)$ $(65, 80)$
$(66, 81)$ $(67, 84)$ $(68, 82)$ $(69, 83)$ $(70, 85)$ $(71, 86)$
$(72, 87)$ $(73, 88)$ $(74, 89)$ $(75, 90)$ $(76, 91)$ $(77, 92)$
$(78, 93)$ $(79, 94)$ $(80, 95)$ $(81, 96)$ $(82, 97)$ $(83, 98)$
$(85, 99)$ $(86, 100)$ $(87, 107)$ $(88, 106)$ $(89, 110)$ $(90, 113)$
$(91, 116)$ $(92, 119)$ $(93, 123)$ $(94, 126)$ $(95, 131)$ $(96, 130)$
$(97, 134)$ $(98, 137)$ $(99, 140)$ $(100, 144)$ $(101, 145)$ $(103, 148)$
$(105, 151)$ $(107, 154)$ $(109, 157)$ $(111, 160)$ $(113, 163)$ $(115, 166)$
$(117, 169)$ $(119, 172)$ $(121, 175)$ $(123, 178)$ $(125, 181)$ $(127, 184)$
$(129, 187)$ $(131, 141)$ $(132, 188)$ $(133, 0)$ $(134, 200)$ $(136, 201)$
$(138, 202)$ $(139, 191)$ $(140, 194)$ $(141, 142)$ $(142, 192)$ $(143, 193)$

$$(145,128)\quad(150,213)\quad(151,214)$$

C The compact companion's four fixed coefficients

The direct integer family does not use this appendix. For the compact positive-domain family, the following finite recipe supplies the actual four coefficients, including the exact seed. The accompanying independent constructor reproduces the stored bytes and their hashes. In this appendix K denotes the port K_0 in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0,\dots,942\} \cup \{983,\dots,992\} \cup \{1003,\dots,1011\} \\ & \cup \{1013,\dots,1031\} \cup \{1063,1064,1095,2195,3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e\in\mathcal{E}\setminus\{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51}-2)(\rho^{4393}+\rho^{5495}) \\ &\quad + \rho^{1013}G_{10}(\rho) + \rho^{1065}G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65}-1)\rho^{8792}G_{13}(\rho) + (2^{19}-1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393}+\rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1+2^{51})G_{943}(x) - 2^{10}\sum_{i=0}^{942} v(t_i)x^i \\ &\quad + \sum_{(b,e,m)\in\mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10,1023,8),(18,1013,9),(27,1003,8), \\ &\quad (35,983,9),(44,1031,1),(45,1022,1),(46,1011,1), \\ &\quad (47,992,1),(48,1095,1),(49,2195,1),(50,3296,1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0,\dots,q-1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j)l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\,\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b = 0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\,\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is:

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSD0Tgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```